

SE200 Gas Compressor Drive (GCD)

Version 1.0

Operations Manual

SOFTWARE VERSION 1.14

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Table of Contents

Keypad (HMI)	3
Power and Motor Terminations	4
Control Wiring.....	5
Initial Power Up	9
Table 1 Startup Guide	16
Main Screen	17
Status Page.....	19
Motor Setup.....	21
I/O Status	22
S/D Debounce & Startup Delays	23
Suction S/D & Control	25
Discharge S/D & Control	26
Suction Sleep/Wake Function	27
Discharge Sleep/Wake Function	28
Trending #1 (Motor Data)	29
Trending #2 (Pressure).....	30
Trending Scales	31
Alarm History	32
Login Pop-up	33
Troubleshooting	34
Faults/Alarms	34

Keypad (HMI)

The main screen of the SE200's HMI includes several buttons and indicators to provide feedback and control of the VFD. An example of the main screen is shown in the figure below.

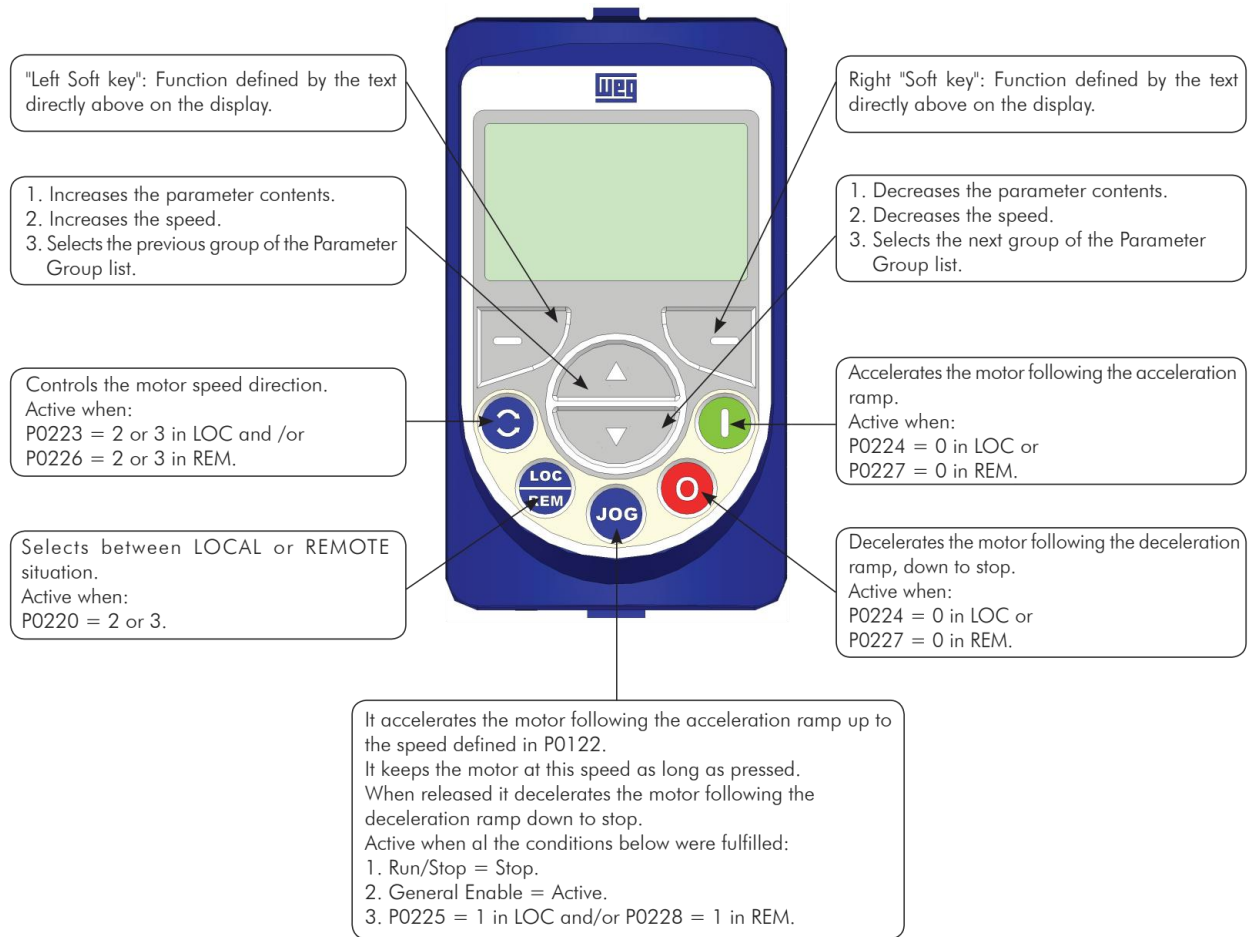


Figure 1 Keypad

Power and Motor Terminations

The SE200 VFD supports both single-phase and three-phase applications. When using it in a single-phase setup, follow these key guidelines to ensure proper operation:

1. Power Input Connections

- Connect **single-phase** power to A/L1 (**Red**) and B/L2 terminals (**Black**). Connect **three-phase** power to A/L1 (**Red**), B/L2 terminals (**Black**) and C/L3 terminals (**Blue**).
- These terminals supply power to the DC bus of the CFW11.

2. Motor Wiring

- Terminate the motor leads at the U, V, and W terminals, located at the bottom right of the VFD.

3. Sizing Requirement

- When using the SE200 in single-phase mode, select a VFD rated at twice the intended motor load.
- This ensures sufficient capacity and reliable performance.

4. Parameter Configuration

- Set P0357 = 0 sec to bypass the phase loss fault (this is required for operation).

Following these steps will help ensure efficient and reliable operation of the SE200 VFD in a single-phase configuration

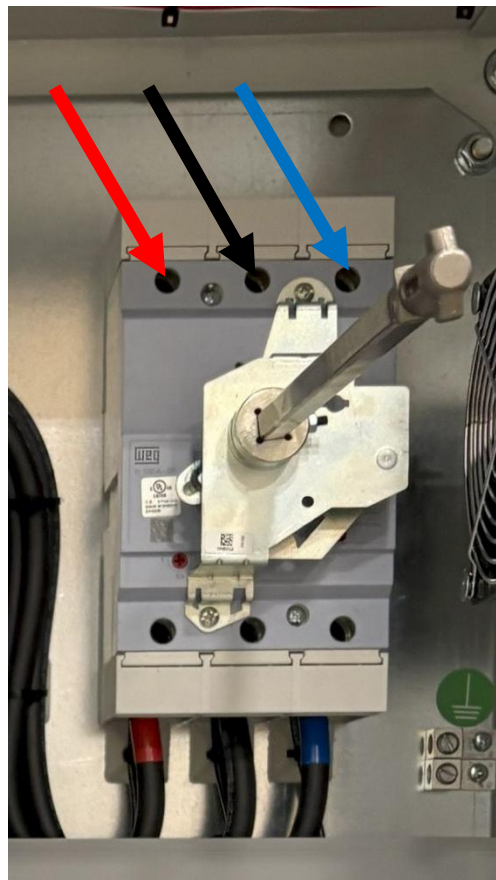


Figure 2 Main Breaker

Control Wiring

For external shutdowns listed below, the control wiring can be connected as outlined below.

1. Jumper Removal

- Remove jumpers only for the specific shutdowns that are required.

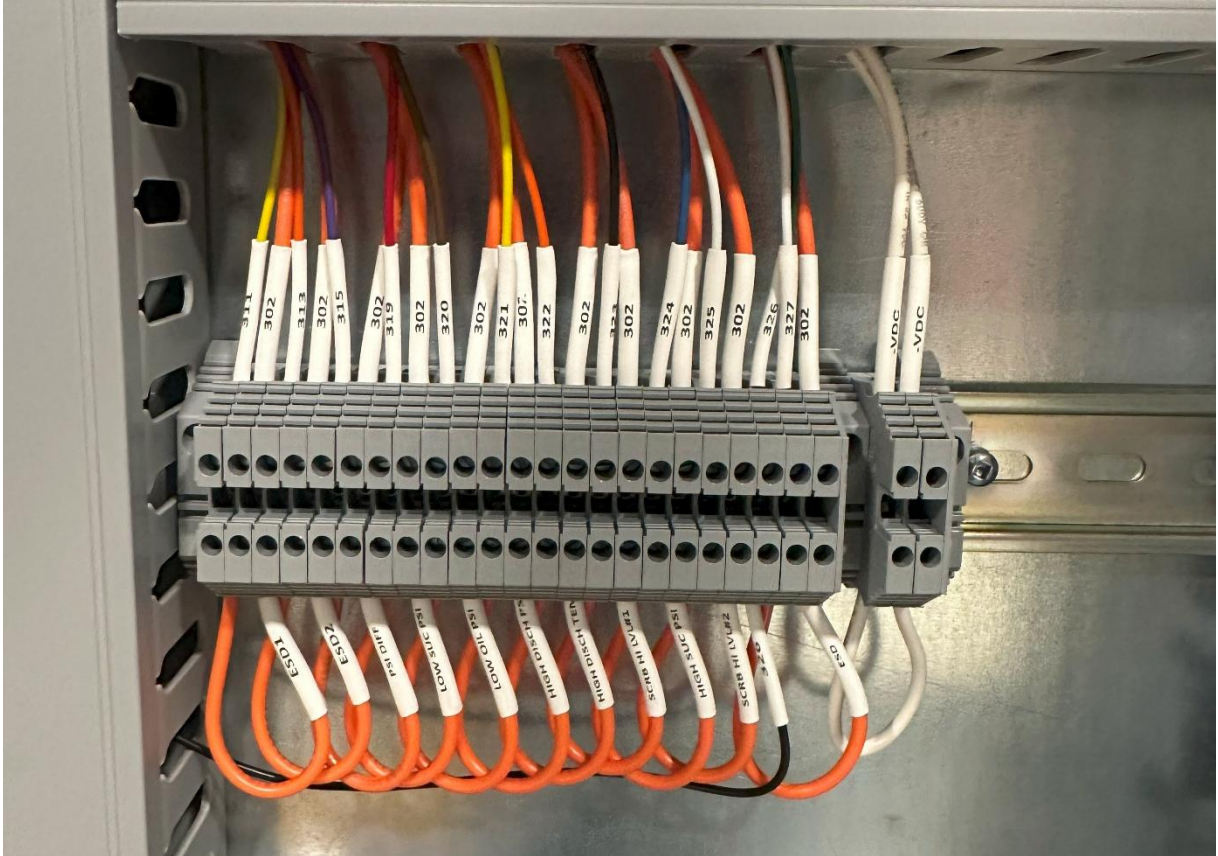


Figure 3 Terminal strip layout

2. Alarm Indications

- The following alarms will appear on the HMI for corresponding shutdowns:
 - A750: Panel ESD
 - A751: High Discharge Pressure-SW
 - A752: High Discharge Temperature
 - A753: Scrubber High Level #1
 - A754: High Suction Pressure
 - A755: Scrubber High Level #2
 - A757: Low Suction Pressure (Switch)
 - A758: Low Oil Pressure
 - A759: Shutdown #1
 - A760: Shutdown #2
 - F761: Pressure Differential Shutdown
 - F762: Low Suction Pressure (Transmitter)
 - F764: Both PID's Enabled
 - A765: Suction PID Sleep
 - A766: Discharge PID Sleep

3. Additional Information

- Refer to the wiring diagrams below for detailed instructions and additional guidance.

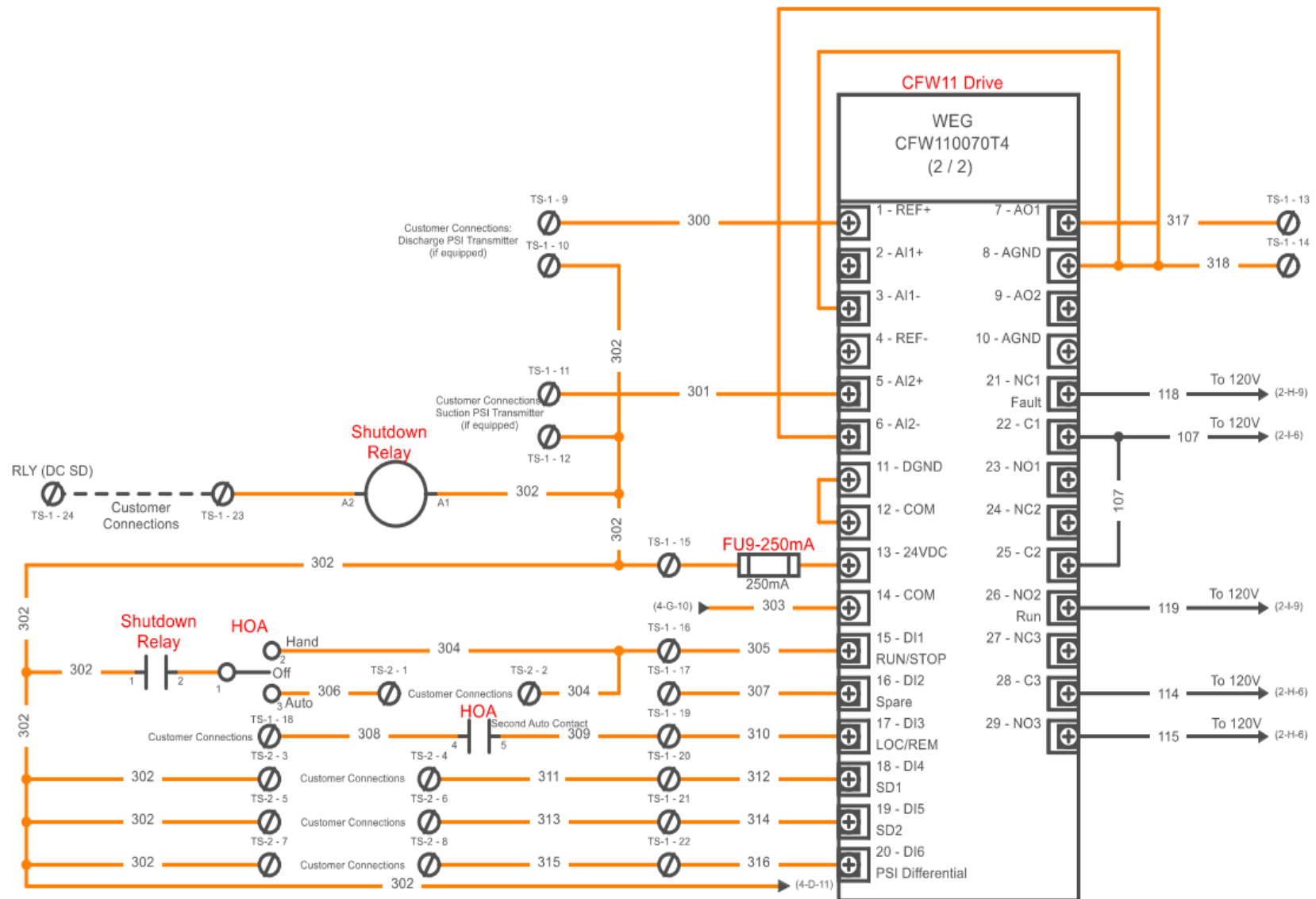


Figure 4 Terminal strip layout

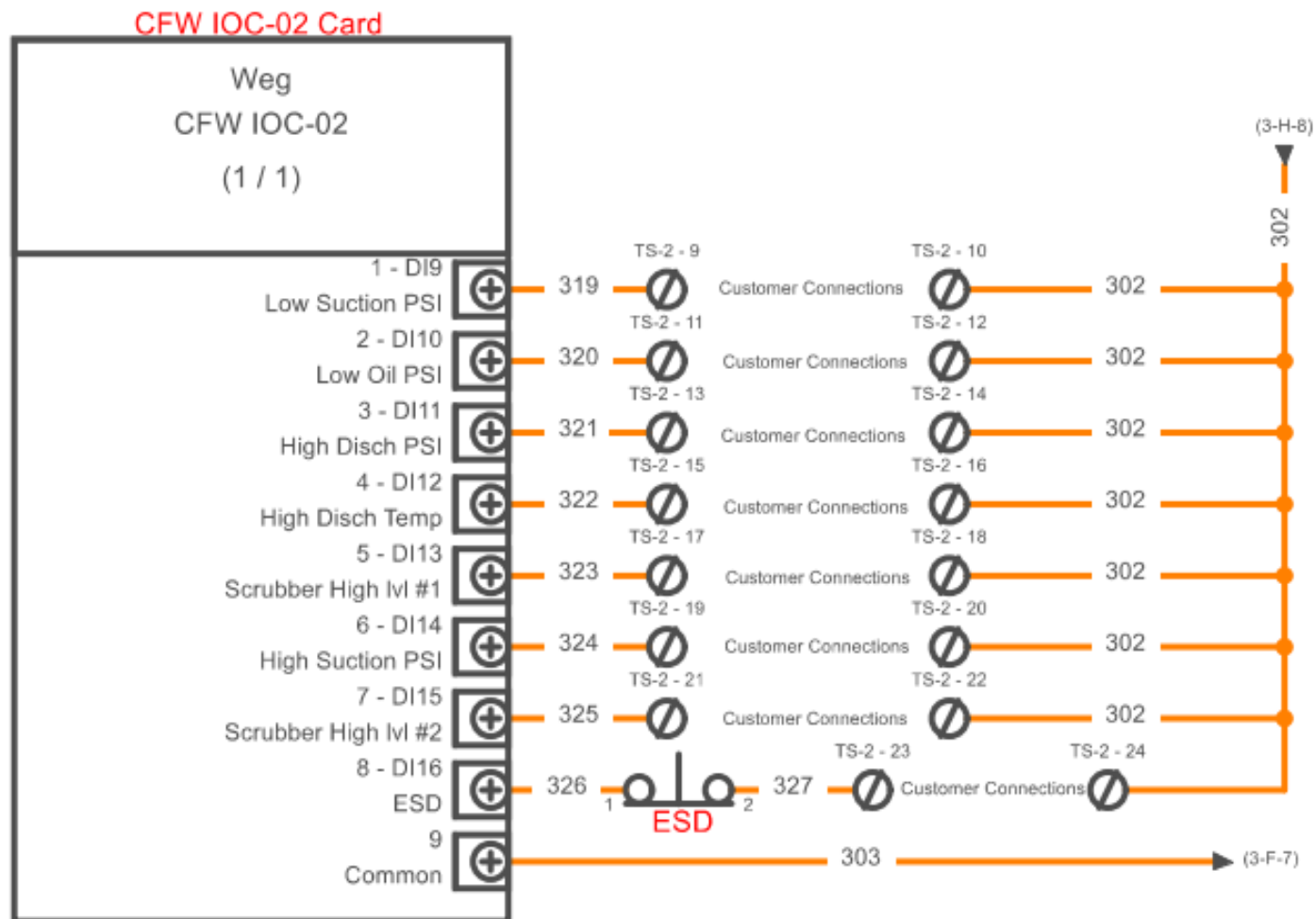


Figure 5 Terminal strip layout

Initial Power Up

When powering up the SE200 for the first time, the field technician must configure the device using the **Field Technician Parameters** section outlined in our startup guide. These parameters are detailed below, and a full copy of the guide is included with every SE200 unit. **Parameter steps #1-#29 are set in SEEL factory and do not need to be set in the field unless otherwise told by SEEL Representative, or in other parts of this manual.**

Steps to Access and Configure Parameters:

1. **Access the Parameter Section**
 - Press **Menu > Select All Parameters**.
 - Use the **Up** and **Down** arrows to navigate to parameter **P0202**.
2. **Complete the Oriented Startup Menu**
 - Starting at Step 1, the SE200 VFD will guide you through the oriented startup steps in **yellow** (Steps 30–37).
 - After completing the auto-tune process, the device will return to the main VFD screen.
3. **Continue Programming**
 - Return to **Menu > All Parameters** to proceed with the programming procedure (Steps 37–85)

PARAMETERS SET AT SEEL FACILITY (Steps #1-29)					
Step#	Parameter #	Parameter Description	Unit	Default Value	Set Value
1	P0000	Access to Parameters	0-9999	0000	0005
2	P0100	Acceleration Time	Sec	20.0	15.0
3	P0101	Deceleration Time	Sec	20.0	15.0
4	P0217	Zero Speed Disable	Disable Magnetization Current	0	1
5	P0220	LOC/REM Selection Source	Dlx	2	4
6	P0221	LOC Reference Select	Keypad-(Transmitter), AI1-(Speed Pot)	0	0
7	P0222	Remote Reference Select	Soft PLC	1	12
8	P0223	LOC FWD/REV	Always FWD	2	0

9	P0224	LOC Run/Stop Selection	Soft PLC	0	5
10	P0225	LOC Jog Selection	Disable	1	0
11	P0226	REM FWD/REV	Always FWD	4	0
12	P0227	REM Run/Stop Selection	Soft PLC	1	5
13	P0228	REM Jog Selection	Disable	2	0
14	P0229	Stop Mode Selection	Coast to Stop	0	1
15	P0231	AI1 Signal Function	PLC Use (Set if using discharge PID control)	0	7
16	P0233	AI1 Signal Type	0-10vdc (0) or 4-20mA (1)	0	0
17	P0236	AI2 Signal Function	PLC Use (Set if using suction PID control)	0	7
18	P0238	AI2 Signal Type	0-10vdc (0) or 4-20mA (1)	0	1
19	P0263	DI1 Function	Run/Stop	1	1
20	P0264	DI2 Function	Not Used	8	0
21	P0265	DI3 Function	LOC/REM	0	9
22	P0266	DI4 Function	PLC Use	0	21
23	P0267	DI5 Function	PLC Use	10	21
24	P0268	DI6 Function	PLC Use	14	21
25	P0275	DO1 Function (RL1)	No Fault	13	13
26	P0276	DO2 Function (RL2)	Run	2	11
27	P0277	DO3 Function (RL3)	Soft PLC	1	28
28	P0320	Fly Start/Ride-Through	Ride-Through	0	3
29	P0353	IGBT's / Air Overtemp Config	Under Temp Disable	0	4
PARAMETERS TO BE SET BY FIELD PERSONNEL (Steps #30-93)					
30	P0317	Oriented Start Up	Yes	0	1
31	P0398	Motor Service Factor	As per motor nameplate		Motor SF
32	P0400	Motor Rated Voltage	As per motor nameplate		Motor Voltage
33	P0401	Motor Rated Current	As per motor nameplate		Motor Current
34	P0402	Motor Rated Speed	As per motor nameplate		Motor RPM
35	P0403	Motor Rated Frequency	As per motor nameplate		Motor Hz
36	P0404	Motor Rated Power	As per motor nameplate		Motor HP/KW
37	P0407	Rated Power Factor	0.5 to 0.99	0.68	Set as per nameplate
38	P0156	Overload Current 100% Speed	0.1 to 1.5 x Inom-ND Example: Motor FLA – 13.2A If SF is 1.0, P0156 should be set to 15.2A	1.05 x Inom-ND	As per motor service factor, adjust if required (SF=1.0 – 1.15 x Inom-ND)

			If SF is 1.15, P0156 should be set to 16.5A		(SF=1.15 – 1.25 x Inom-ND)
39	P0157	Overload Current 50% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.9 x Inom-ND	As per motor service factor, adjust if required
40	P0158	Overload Current 5% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.65 x Inom-ND	As per motor service factor, adjust if required
41	P0159	Motor Thermal Class	Class 5,10,15,20,25,30,35,40,45	1 – Class 10	3 – Class 20
42	P0296	Line Rated Voltage	0-6 (Can be adjusted if overvoltage or undervoltage trips occur)	6- (575)	Set as per input voltage
43	P0340	Auto Reset Time (VFD Faults)	0-3600s	0	Set to desired reset time
44	P0357	Line Phase Loss Time	0-60sec	3 sec	Set to 0 seconds
45	P1001	Soft PLC Command	Ensure this is set to “Run Program”	0	1
46	P1010	Suction Pressure (read only parameter, if using suction transmitter set P0205 to 21 (Soft PLC P1010) to display actual Suction Pressure on HMI	PSI	0	Read Only
47	P1011	Discharge Pressure (read only parameter, if using discharge transmitter set P0205 to 22 (Soft PLC P1011) to display actual Discharge Pressure on HMI	PSI	0	Read Only
48	P1012	Shutdown #1 Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs

49	P1013	Shutdown #2 Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
50	P1014	Pressure Differential Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
51	P1015	Low Suction Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
52	P1016	Low Oil Pressure Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
53	P1017	High Discharge Pressure Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
54	P1018	High Discharge Temp Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
55	P1019	Scrubber High Level #1 Shutdown Debounce (High Level #1 is a Shutdown Only)	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs

56	P1020	High Suction Pressure Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
57	P1021	Scrubber High Level #2 Shutdown Debounce (High Level #2 is a Shutdown and controls the liquid run pump)	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
58	P1022	Pressure Differential Start Delay	Sec	0	Set to desired time before shutdown is active, timer starts after the start pushbutton is pressed
59	P1023	Liquid Pump Run Time (Will run for set time when High Level #2 is tripped)	Sec	0	Set to desired time for liquids pump to run, a run condition is initiated when Scrubber High Level #2 is triggered
60	P1024	Suction Pressure Transmitter Range (Ex: maximum unit the transmitter is ranged at for 100% scale)	PSI (Note when configuring a transmitter with a negative range you must take the total range of the transmitter divided by the negative value e.g.; -15 to 30 transmitters would be $45/15 = 33.3\%$ this number must be entered into P0239 as a negative offset.)	0	maximum unit the transmitter is ranged at for 100% scale
61	P1025	Suction Pressure Transmitter Low Shutdown Setpoint (If using transmitter for shutdown digital input will need to be jumpered)	PSI	0	Set to desired low suction shutdown setpoint

62	P1026	Suction Pressure Transmitter Low Suction Shutdown Pressure Scale (Read only)	n/a	0	
63	P1027	Suction Pressure Transmitter Shutdown Enable	n/a	0	Set to 1 to enable transmitter to initiate shutdown
64	P1028	Discharge Pressure Transmitter Range (Ex: maximum unit the transmitter is ranged at for 100% scale)	PSI	0	maximum unit the transmitter is ranged at for 100% scale
65	P1029	Discharge Pressure Transmitter High Shutdown Setpoint (If using transmitter for shutdown digital input will need to be jumpered)	PSI	0	Set to desired high discharge shutdown setpoint
66	P1030	Discharge Pressure Transmitter High Discharge Shutdown Pressure Scale (Read only)	n/a	0	
67	P1031	Discharge Pressure Transmitter Shutdown Enable	n/a	0	Set to 1 to enable transmitter to initiate shutdown
68	P1032	Low Oil Pressure Start Delay (time after start is initiated before shutdown is active)	Sec	0	Set to desired time before shutdown is active after start
69	P1033	Low Suction Pressure Start Delay (time after start is initiated before shutdown is active)	Sec	0	Set to desired time before shutdown is active after start
70	P1034	HMI Start (read only)	n/a	n/a	n/a
71	P1035	HMI Stop (read only)	n/a	n/a	n/a

72	P1040	Suction Pressure Transmitter PID Setpoint (set to pressure in PSI the PID will control at, AI2)	PSI	0	Set to desired PID setpoint
73	P1041	Discharge Pressure Transmitter PID Setpoint (set to pressure in PSI the PID will control at, AI2)	PSI	0	Set to desired PID setpoint
74**	P1042	High Suction Pressure Transmitter Shutdown Enable	n/a	0	Set to 1 to enable the high suction pressure shutdown via TX
75**	P1043	High Suction Pressure Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
76**	P1044	High Suction Pressure Shutdown Setpoint	PSI	0	Set to desired shutdown setpoint
77**	P1049	Low Discharge Pressure Transmitter Shutdown Enable	n/a	0	Set to 1 to enable the Low Discharge pressure shutdown via TX
78**	P1050	Low Discharge Pressure Shutdown Debounce	Sec	0	Set to desired time that shutdown needs to be enabled before a VFD fault occurs
79**	P1051	Low Discharge Pressure Shutdown Setpoint	PSI		Set to desired shutdown setpoint
80**	P1056	Low Discharge Pressure Start Delay	Sec	0	Set to desired time before shutdown is active, timer starts after the start pushbutton is pressed
81	P1052	Suction Pressure PID1 Enable (alarm will occur if more than 1 PID block is enabled)	n/a	0	Set to 1 to enable PID
82	P1053	Discharge Pressure PID2 Enable (alarm will occur if more than 1 PID block is enabled)	n/a	0	Set to 1 to enable PID
83	P1057	Power loss Restart	n/a	0	Set to 1 to enable

84	P1054	Suction Sleep/Wake Enable	n/a	0	Set to 1 to enable function
85	P1055	Discharge Sleep/Wake Enable	n/a	0	Set to 1 to enable function
86	P1036	(AI1) Discharge Sleep Setpoint (if the discharge pressure goes above the sleep value the drive will go into "sleep mode")	PSI	0	Set to desired pressure to put drive to "sleep"
87	P1037	Discharge Sleep Debounce	Sec	0	Set to desired time before drive goes to "sleep"
88	P1038	(AI1) Discharge Wake Setpoint (if the discharge pressure point goes below the setpoint the VFD will "wake up")	PSI	0	Set to desired pressure to put drive to "wake up"
89	P1039	Discharge Sleep Wake Debounce	Sec	0	Set to desired time before drive "wakes up"
90	P1045	(AI2) Suction Sleep Setpoint (if the discharge pressure goes below the sleep value the drive will go into "sleep mode")	PSI	0	Set to desired pressure to put drive to "sleep"
91	P1046	Suction Sleep Debounce	Sec	0	Set to desired time before drive goes to "sleep"
92	P1047	(AI2) Suction Wake Setpoint (if the discharge pressure point goes above the setpoint the VFD will "wake up")	PSI	0	Set to desired pressure to get drive to "wake up"
93	P1048	Suction Sleep Wake Debounce	Sec	0	Set to desired time before drive "wakes up"

Table 1 Startup Guide

****Parameters marked with asterisk are not available on the current version of the GCD HMI software and only accessible via VFD HMI.**

Main Screen

The main screen of the GCD's HMI is the central hub of navigation when programming the GCD, this main screen contains every button to navigate you through your GCD setup.

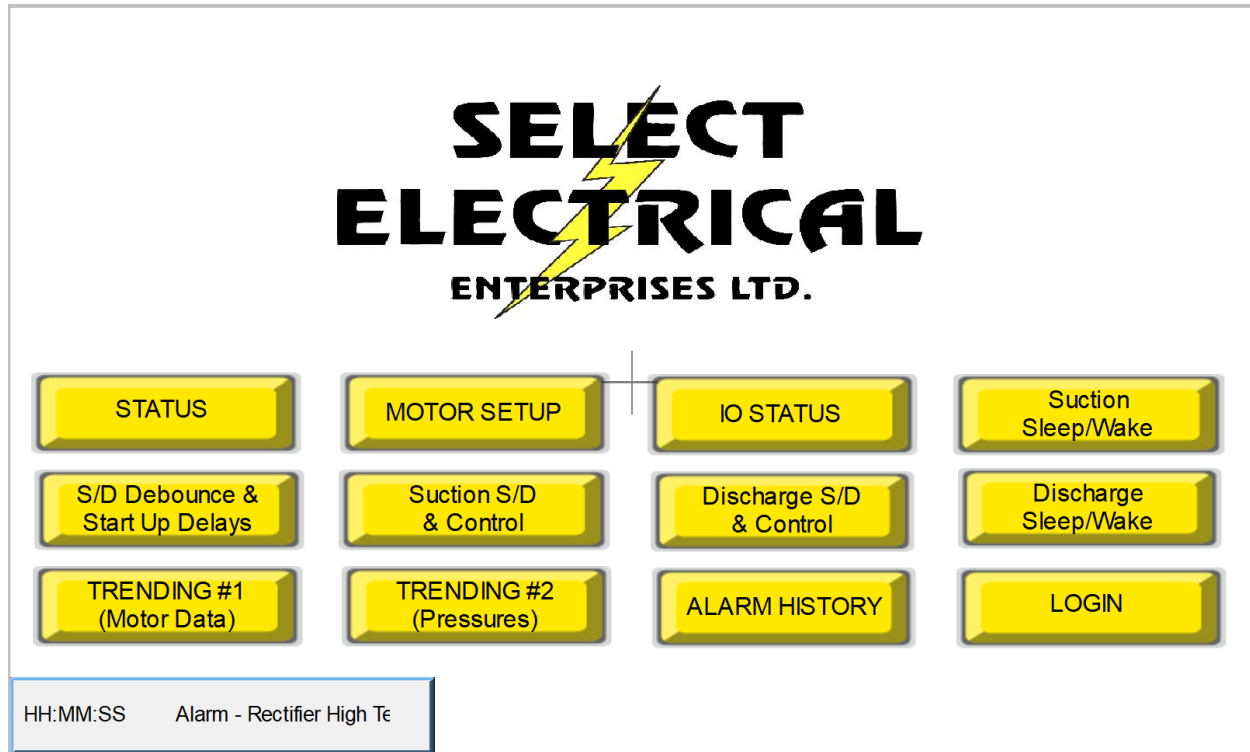


Figure 6 Main Screen

The following list describes the function of each button:

- **STATUS** – The Status button, when pressed, will take you to the status page, this is the main operation page of the SE200 GCD.
- **MOTOR SETUP** – The motor setup button, when pressed, will take you to the motor configuration menu where you can enter the motor data and auto tune the motor.
- **I/O STATUS** – The I/O status button, when pressed, will take you to the I/O status page, this will be a helpful tool for troubleshooting faults, or issues with the GCD in the future.
- **S/D DEBOUNCE & START UP DELAYS** – The S/D Debounce/Start up delay's button, when pressed, will take you to the s/d and debounce menu, this menu is where you are able to set up s/d specific debounces and start up delays.
- **SUCTION S/D & CONTROL** – The Suction s/d and control button, when pressed, will take you to the Suction shutdown and control menu, this is where you will enable your PID function, set up your suction transmitter settings, and configure associated shutdowns.

- **DISCHARGE S/D & CONTROL** – The Discharge s/d and control button, when pressed, will take you to the Discharge shutdown and control menu, this is where you will enable your PID function, set up your discharge transmitter settings, and configure associated shutdowns.
- **SUCTION SLEEP/WAKE** – The Suction sleep/wake button, when pressed, will take you to the Suction sleep/wake menu, in this menu you will configure your sleep/wake function for suction pressure.
- **DISCHARGE SLEEP/WAKE** – The Discharge sleep/wake button, when pressed, will take you to the Discharge sleep/wake menu, in this menu you will configure your sleep/wake function for discharge pressure.
- **TRENDING #1 (MOTOR DATA)** – The Trending #1 button, when pressed, will take you to the Motor data trending page, from this page you will be able to see the trend of motor data over time.
- **TRENDING #2 (PRESSURES)** – The Trending #2 button, when pressed, will take you to the Pressure trending page, from this page you will be able to see the trend of pressure over time.
- **ALARM HISTORY** – When pressed, this button opens the alarm history page.
- **LOGIN** – When pressed, this button opens the log in menu pop-up menu, from here you will be able to log in to the Technician level. See login page for password

Note: Any menu in orange is only available after technician mode is enabled through the login pop-up menu.

Status Page

The Status page is the main hub for operation of the SE200 GCD from this screen the operator will be able to start, stop and reset faults/alarms, set their suction or discharge PID setpoints, view suction and discharge pressures, along with a handful of motor monitoring parameters and any current faults/alarms that may be active. From this page we can also access the main menu, and alarm history pages.

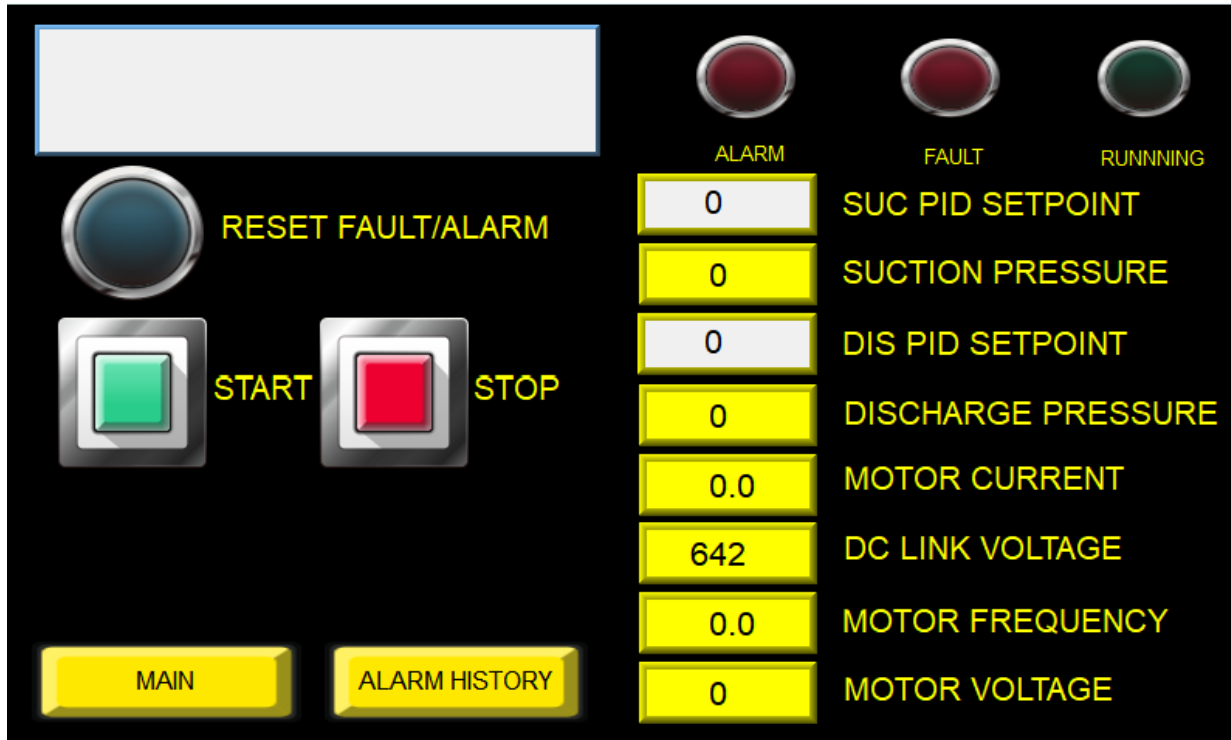


Figure 7 Status page

- **START** – The Start Pumping Unit button, when pressed, will signal the GCD to start the unit. If the unit is in a faulted state, the fault must be reset before the unit will be able to start.
- **STOP** – The Stop Pumping Unit button, when pressed, will signal the GCD to stop the unit.
- **MAIN** – Pressing the Main button will take you to the main menu.
- **ALARM HISTORY** – Pressing the Alarm history button will take you to the alarm history page.
- **SUC PID SETPOINT** – In this parameter you will enter the target suction pressure that you want the GCD to maintain.
SUCTION PRESSURE – This parameter is the active suction pressure value from the transmitter
- **DIS PID SETPOINT** – In this parameter you will enter the target discharge pressure that you want the GCD to maintain.
DISCHARGE PRESSURE – This parameter is the active discharge pressure value from the transmitter

- **MOTOR CURRENT** - This parameter, is the active motor current value, also displayed in the trending page #1
- **DC LINK VOLTAGE**- This parameter, is the active DC link voltage value, also displayed in the trending page #1
- **MOTOR FREQUENCY**- This parameter, is the active motor output frequency value, also displayed in the trending page #1
- **MOTOR VOLTAGE**- This parameter, is the active motor output voltage value, also displayed in the trending page #1

Motor Setup

The motor setup page provides the technician a simplified motor setup with all parameters required laid out on one page.

1.00	Motor Service Factor (1.0-1.50)	Motor Rated Power
67.0	Motor Rated Efficiency (50.0-99.9%)	0.5HP 0.37KW
460	Motor Rated Voltage	Line Rated Voltage
0.5	Motor Rated Current	440-460V
1750	Motor Rated Speed	Type of Control
60	Motor Rated Hz	7 = Sensorless PM
0.68	Motor Rated Power Factor	Run Self Tuning
20.0	Acceleration Time	0 = No
20.0	Deceleration Time	MAIN
90	Minimum Speed	STATUS
1800	Maximum Speed	

Figure 8 Motor setup page

- **MOTOR PARAMETERS** – Enter the nameplate data from the motor into the corresponding parameter selection above
- **ACCELERATION TIME** – The time it takes for the GCD to accelerate to the maximum speed lineally from 0
- **DECELERATION TIME** – The time it takes for the GCD to decelerate from the maximum speed lineally to 0
- **MINIMUM SPEED** – The minimum speed the GCD will run. (typically, set to minimum speed required to lubricate system ask pump tech for specific speed)
- **MAXIMUM SPEED** – The maximum speed the GCD can operate the motor at. (typically, around 1200 rpm ask pump tech for specific setpoint)

I/O Status

The I/O Status page shows a breakdown of the I/O status for the GCD along with the incoming pressures displayed as a percentage value. This page is primarily for technician use during troubleshooting. When the lights are **Green**, it means that the inputs are satisfied (+24VDC on terminal) when they are **Red**, it means that the input is off (0VDC on terminal). The kPA and PSI sliders change the units on the main page.

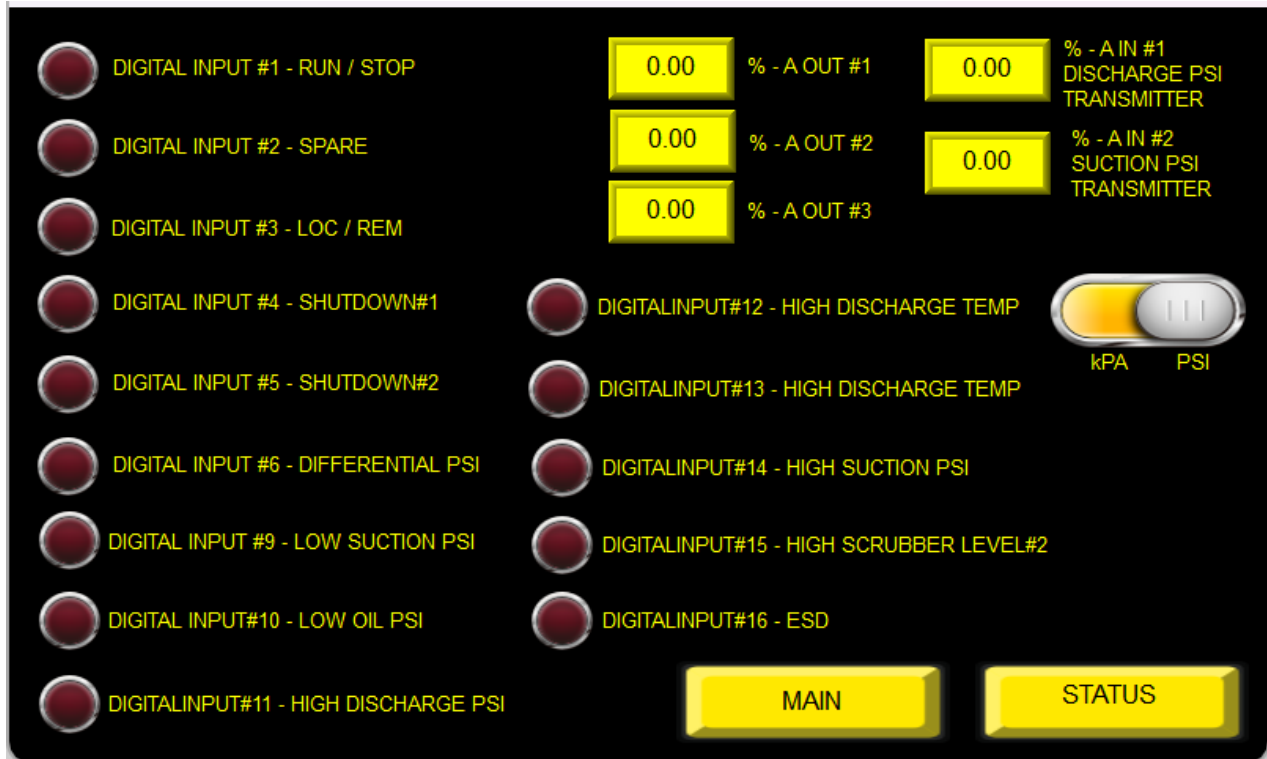


Figure 9 I/O Status page

S/D Debounce & Startup Delays

The S/D Debounce & Start up Delays page is where the technician can set and adjust shutdown debounce(delay) timers and start up delay timers for the specific application. Note this page is only accessible when technician mode is enabled.

0	Suction Pressure	0	High Suct Press Debounce (Sec)
0	Discharge Pressure	0	High Level Scrubber#2 Debounce (Sec)
0	Shutdown#1 Debounce (Sec)	0	PSI Differential Start Delay (Sec)
0	Shutdown#2 Debounce (Sec)	0	Liquid Pump Run Time (Sec)
0	PSI Differential Debounce (Sec)	0	Low Oil Press Start Delay (Sec)
0	Low Suct Press Debounce (Sec)	0	Low Suct Press Start Delay (Sec)
0	Low Oil Press Debounce (Sec)		
0	High Disch Press Debounce (Sec)		
0	High Disch Temp Debounce (Sec)		
0	High Level Scrubber#1 Debounce (Sec)		

MAIN

STATUS

Figure 10 S/D Debounce & Startup page

- **SHUTDOWN DEBOUNCE #1** – This parameter allows the user to set a shut down delay for shutdown #1 if the shutdown stays in an “open state” after the set time the unit will fault.
- **SHUTDOWN DEBOUNCE #2** – This parameter allows the user to set a shut down delay for shutdown #2 if the shutdown stays in an “open state” after the set time the unit will fault.
- **PSI DIFFERENTIAL DEBOUNCE** – This parameter allows the user to set a time delay for the pressure differential s/d if the unit stays in high pressure differential after the set time it will fault.
- **LOW SUCTION PRESSURE DEBOUNCE** – This parameter allows the user to set a time delay for the low suction pressure s/d if the unit stays in low pressure after the set time it will fault.
- **LOW OIL PRESSURE DEBOUNCE** – This parameter allows the user to set a time delay for the low oil pressure s/d if the unit stays in low pressure after the set time it will fault.
- **HIGH DISCHARGE PRESSURE DEBOUNCE** – This parameter allows the user to set a time delay for the high discharge pressure s/d if the unit stays in high pressure after the set time it will fault.
- **HIGH DISCHARGE TEMP DEBOUNCE** – This parameter allows the user to set a time delay for the high discharge temp s/d if the unit stays in high temp after the set time it will fault.

- **HIGH LEVEL SCRUBBER #1 DEBOUNCE** – This parameter allows the user to set a time delay for the high scrubber level #1 s/d if the unit stays in high level after the set time it will fault.
- **HIGH SUCTION PRESSURE DEBOUNCE** – This parameter allows the user to set a time delay for the high suction pressure s/d if the unit stays in high pressure after the set time it will fault.
- **HIGH LEVEL SCRUBBER #2 DEBOUNCE** – This parameter allows the user to set a time delay for the high scrubber level #2 s/d if the unit stays in a high level after the set time it will fault.
- **PSI DIFFERENTIAL START DELAY** – This parameter allows the technician to set a start up delay to allow the unit to start the pump with the pressure above the S/D set limit for a set amount of time so that the GCD can lower the pressure differential.
- **LIQUIDS PUMP RUN TIME** – This sets a time for the liquids pump to run for.
- **LOW OIL PRESSURE START DELAY** – This parameter allows the technician to set a start up delay to allow the unit to start the pump with the oil pressure below the S/D set limit for a set amount of time so that the GCD can build oil pressure.
- **LOW SUCTION PRESSURE START DELAY** – This parameter allows the technician to set a start up delay to allow the unit to start the pump with the suction pressure below the S/D set limit for a set amount of time so that the GCD can build suction pressure.

Suction S/D & Control

The Suction S/D & Control menu is for the technician to enable the suction PID, set up the suction transmitter range, the suction pressure setpoint if using the suction transmitter as the s/d, and from this menu you can also set the debounce, and suction PID setpoint. **NOTE: BOTH SUCTION AND DISCHARGE PIDS CANNOT BE ENABLED AT THE SAME TIME!!!**

0.0	Suction Pressure Transmitter Range	Low Suction Pressure Transmitter S/D Enable
0.0	Low Suction Shutdown Setpoint	
0	Low Suction Shutdown Debounce (Sec)	High Suction Pressure Transmitter S/D Enable
0.0	High Suction Shutdown Setpoint	
0	High Suction Shutdown Debounce (Sec)	Suction Pressure PID Enable
0.0	Suction Pressure PID Setpoint	
0.0	Suction Pressure	MAIN STATUS

Figure 11 Suction S/D & Control Menu page

- **Suction pressure max range** – This is the max range of the suction pressure transmitter. (Note when configuring a transmitter with a negative range you must take the total range of the transmitter divided by the negative value eg; -15 to 30 transmitters would be $45/15 = 33.3\%$ this number must be entered into P0239 as a negative offset.)
- **Low Suction pressure S/D setpoint** – This is the S/D setpoint of the suction pressure transmitter, only active when the Low Suction Pressure TX Enabled slider is on. (Note: the jumper must be installed for the suction pressure s/d switch for this to work properly.)
- **Low Suction pressure S/D debounce** – this is the delay time before the unit will go down on low suction.
- **High Suction pressure S/D setpoint** – This is the S/D setpoint of the suction pressure transmitter, only active when the High Suction Pressure TX Enabled slider is on.
- **High Suction pressure S/D debounce** – this is the delay time before the unit will go down on high suction.
- **Suction pressure PID setpoint** – this is the target suction pressure you want the SE200 GCD to maintain.
- **Suction Pressure** – this is the real time suction pressure in the specified unit.
- **Low Suction Pressure TX Enable** – when this slider is on the suction transmitter will act as the low suction shutdown as opposed to the suction pressure switch. The suction s/d setpoint is then programmed by the above parameter.
- **High Suction Pressure TX Enable** – when this slider is on the suction transmitter will act as the high suction shutdown. The suction s/d setpoint is then programmed by the above parameter.
- **Suction PID Enable** – when this slider is on the suction PID will be activated. **NOTE: BOTH SUCTION AND DISCHARGE PIDS CANNOT BE ENABLED AT THE SAME TIME!!!**

Discharge S/D & Control

The Discharge S/D & Control menu is for the technician to enable the suction PID, set up the discharge transmitter range, the discharge pressure setpoint if using the discharge transmitter as the s/d, and from this menu you can also set the debounce, and discharge PID setpoint. **NOTE: BOTH SUCTION AND DISCHARGE PIDS CANNOT BE ENABLED AT THE SAME TIME!!!**

0.0	Discharge Pressure Transmitter Range	High Discharge Pressure Transmitter S/D Enable
0.0	High Discharge Shutdown Setpoint	☒
0	High Discharge Shutdown Debounce (Sec)	Low Discharge Pressure Transmitter S/D Enable
0.0	Low Discharge Shutdown Setpoint	☒
0	Low Discharge Shutdown Debounce (Sec)	Discharge Pressure PID Enable
0	Low Discharge Start Up Delay (Sec)	☒
0.0	Discharge Pressure PID Setpoint	
0.0	Discharge Pressure	MAIN STATUS

Figure 12 Discharge S/D & Control page

- **Discharge pressure max range** – This is the max range of the discharge pressure transmitter.
- **Discharge pressure S/D setpoint** – This is the S/D setpoint of the discharge pressure transmitter, only active when the Suction Pressure TX Enabled slider is on. **(Note: the jumper must be installed for the discharge pressure s/d switch for this to work properly.)**
- **High Discharge pressure S/D debounce** – this is the delay time before the unit will go down on high discharge.
- **Discharge pressure S/D setpoint** – This is the S/D setpoint of the low discharge pressure transmitter.
- **Low Discharge pressure S/D debounce** – this is the delay time before the unit will go down on low discharge.
- **Low Discharge pressure startup delay** – this is the time set to bypass the low discharge alarm to allow the discharge line to build pressure.
- **Discharge pressure PID setpoint** – this is the target discharge pressure you want the SE200 GCD to maintain.
- **Discharge Pressure** – this is the real time discharge pressure in the specified unit.
- **High Discharge Pressure TX Enable** – when this slider is on the discharge transmitter will act as the high discharge shutdown as opposed to the discharge pressure switch. The discharge s/d setpoint is then programmed by the above parameter.
- **Low Discharge Pressure TX Enable** – when this slider is on the discharge transmitter will enable the low discharge pressure shutdown.
- **Discharge PID Enable** – when this slider is on the discharge PID will be activated. **NOTE: BOTH SUCTION AND DISCHARGE PIDS CANNOT BE ENABLED AT THE SAME TIME!!!**

Suction Sleep/Wake Function

The Suction Sleep/Wake Function page allows the technician to enable/disable and configure the suction sleep/wake function for the SE200 GCD.

5	Suction Sleep Setpoint (PSI)	Suction Sleep/Wake Enable ☒
2	Suction Sleep Debounce (Sec)	
3	Suction Wake Setpoint (PSI)	
5	Suction Wake Debounce (Sec)	
0	Suction Pressure (PSI)	
		MAIN STATUS

Figure 13 Suction Sleep/Wake page

- **Suction sleep setpoint** – When the suction pressure goes **below** the setpoint the unit will enter “sleep mode”. While in sleep mode all alarms are still active and work as normal, however the drive will not run. This is ideal in applications where you want the unit to stop running before it reaches the low suction shutdown and start back up automatically once it passes the wake setpoint.
- **Suction sleep debounce** – This is the time delay before the unit goes to sleep after going below the setpoint.
- **Suction wake setpoint** – When the suction pressure rises **above** the setpoint the unit will “wake up” from sleep mode and enter a normal operating mode.
- **Suction wake debounce** – This is the time delay before the unit wakes up after the pressure increases above the setpoint.
- **Suction Pressure** – this is the real time suction pressure in the specified unit.

Discharge Sleep/Wake Function

The Discharge Sleep/Wake Function page allows the technician to enable/disable and configure the discharge sleep/wake function for the SE200 GCD.

5	Discharge Sleep Setpoint (PSI)	
6	Discharge Sleep Debounce (Sec)	
5	Discharge Wake Setpoint (PSI)	Discharge Sleep/Wake Enable
3	Discharge Wake Debounce (Sec)	☒
0	Discharge Pressure (PSI)	

MAIN

STATUS

Figure 14 Discharge Sleep/Wake page

- **Discharge sleep setpoint** – When the discharge pressure goes **above** the setpoint the unit will enter “sleep mode”. While in sleep mode all alarms are still active and work as normal, however the drive will not run. This is ideal in applications where you want the unit to stop running before it reaches the high discharge shutdown and start back up automatically once it passes the wake setpoint.
- **Discharge sleep debounce** – This is the time delay before the unit goes to sleep after going above the setpoint.
- **Discharge wake setpoint** – When the discharge pressure drops below the setpoint the unit will “wake up” from sleep mode and enter a normal operating mode.
- **Discharge wake debounce** – This is the time delay before the unit wakes up after the pressure drops below the setpoint.
- **Discharge Pressure** – this is the real time discharge pressure in the specified unit.

Trending #1 (Motor Data)

The Trending page shows the motor voltage, frequency, current and torque over time, this trend is scrollable so you can go back in time to see the data on a previous day, it is also available to download to a pc via the USB stick on the back.

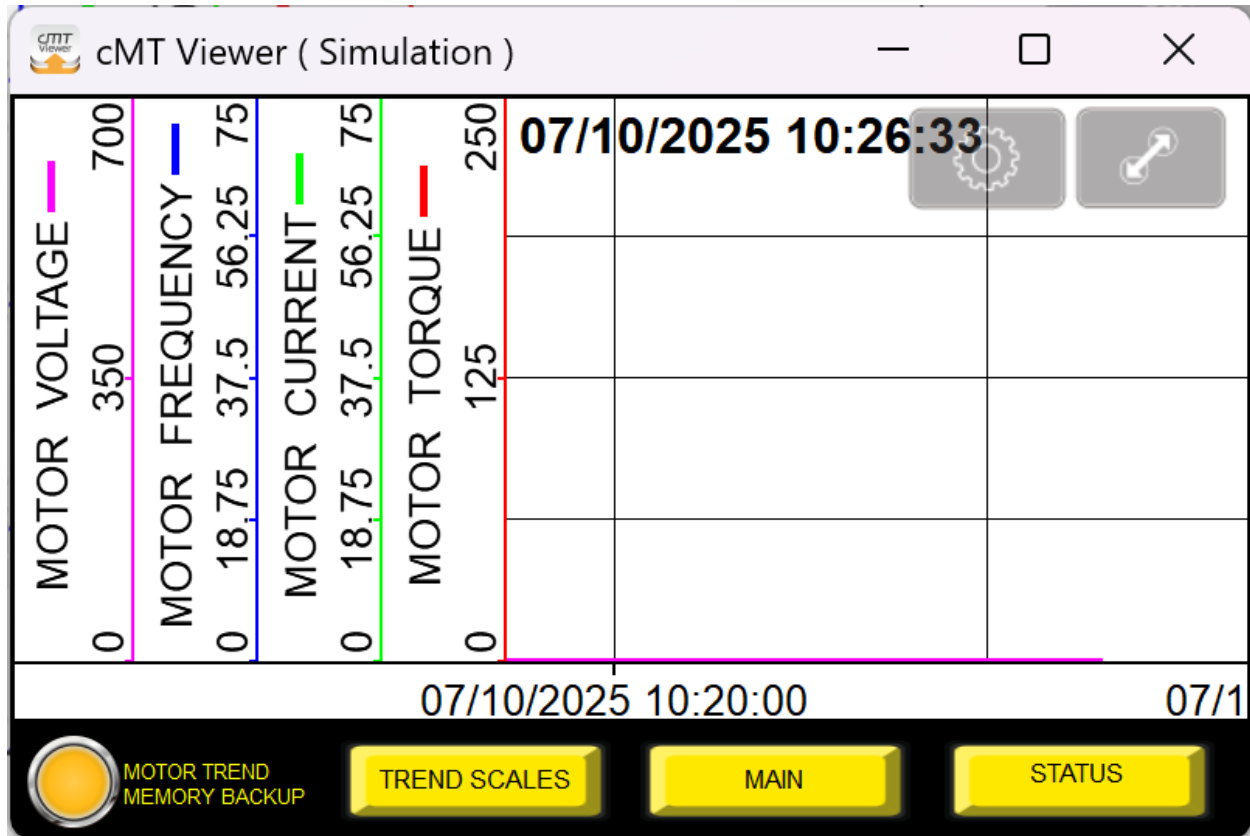


Figure 15 Trending #1 page

Trending #2 (Pressure)

The Trending page shows the suction and discharge pressures over time, this trend is scrollable so you can go back in time to see the data on a previous day, it is also available to download to a pc via the USB stick on the back.

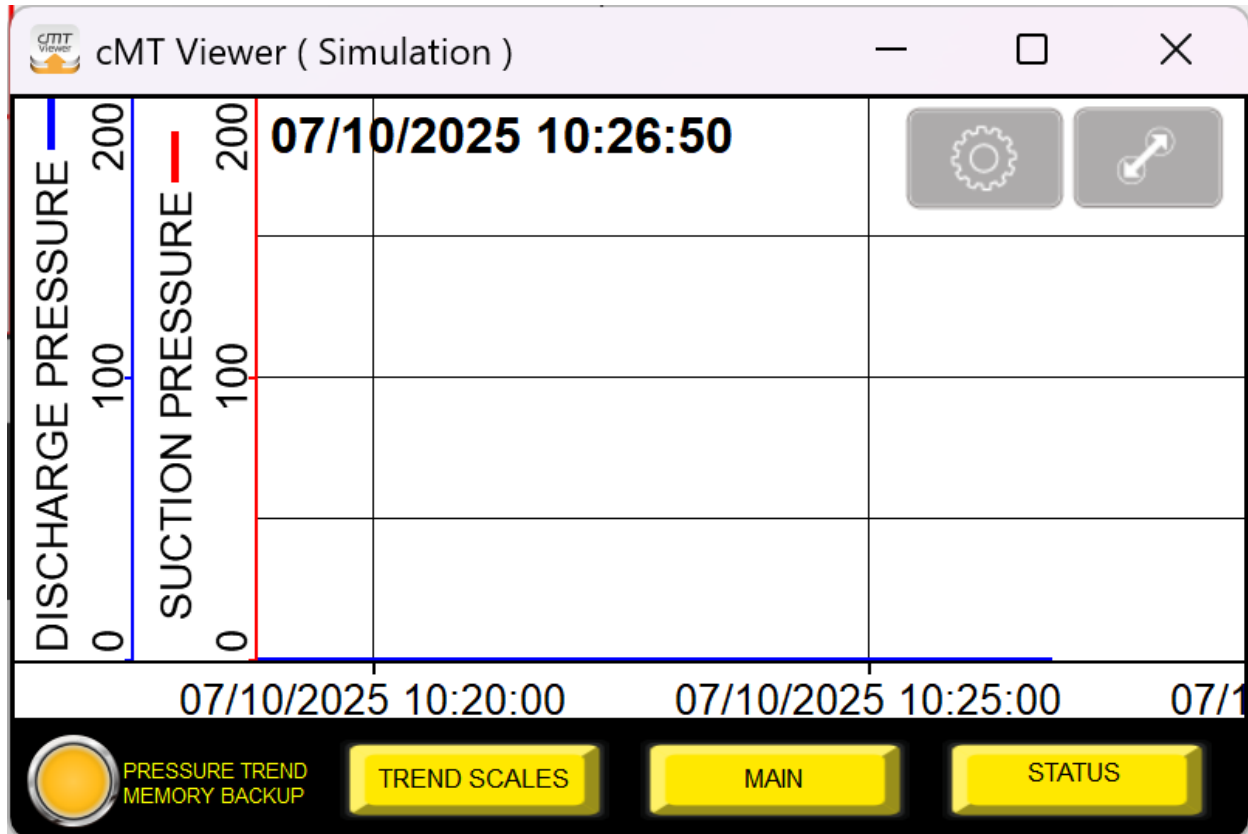


Figure 16 Trending #2 page

Trending Scales

The Trending scales page allows the user to adjust the scaling based off there ranges of the suction/discharge pressure transmitters and the motor current. **Note: This page is only accessible from the Trending #1 and Trending #2-page menus.**

The screenshot displays the 'Trending Scales' interface on a black background. On the left, there are six white input fields with yellow borders, each containing a numerical value. To the right of each field is a yellow label. On the right side of the screen, there are four yellow navigation buttons stacked vertically.

Input Value	Label
-1	Suction PSI Min Scale
100	Suction PSI Max Scale
0	Discharge PSI Min Scale
2	Discharge PSI Max Scale
0	Motor Current Min Scale
75	Motor Current Max Scale

Navigation Buttons:

- TRENDING #1 (Motor Data)
- TRENDING #2 (Pressures)
- MAIN
- STATUS

Figure 17 Trending Scales page

Alarm History

The Alarms page contains several elements for keeping track of the faults and alarms of the GCD. The alarm history page is also able to be downloaded to a USB to download the history.

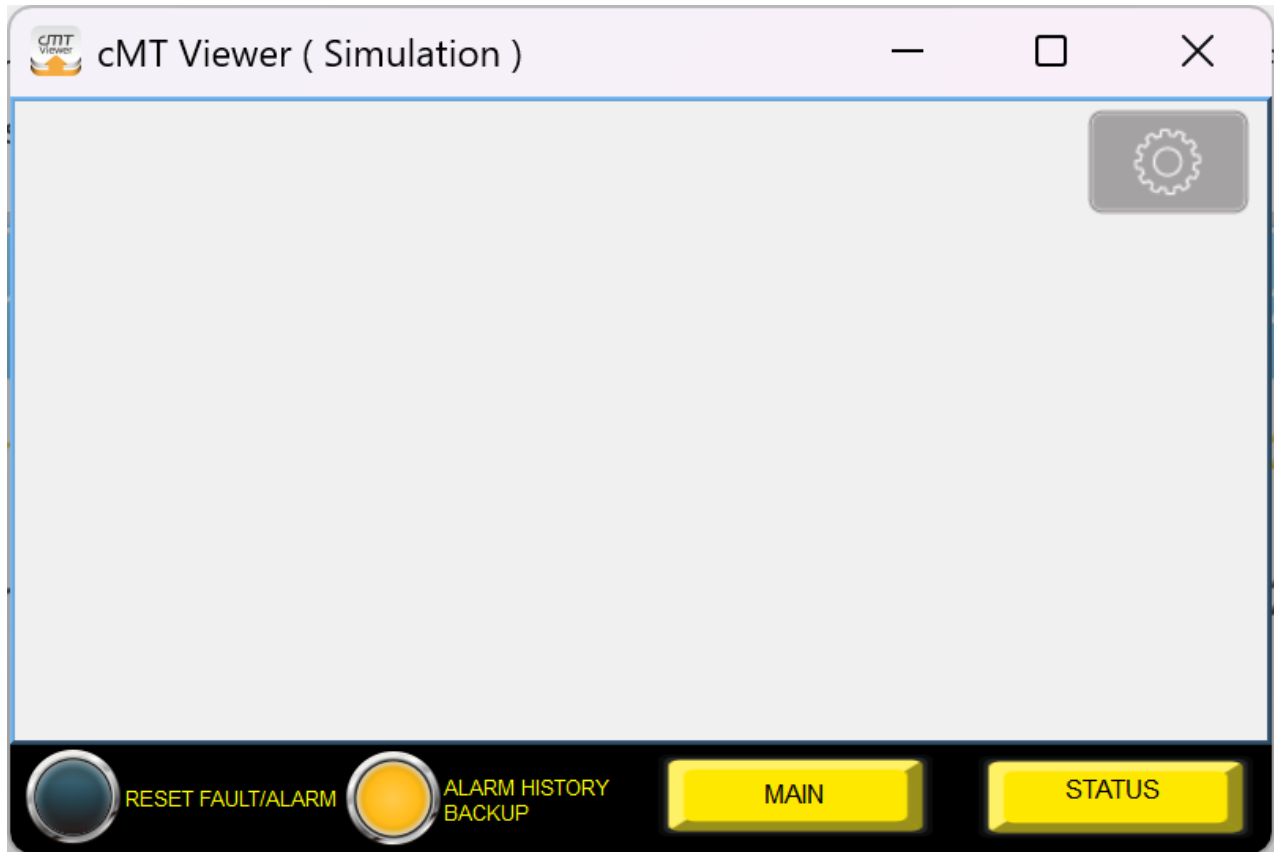
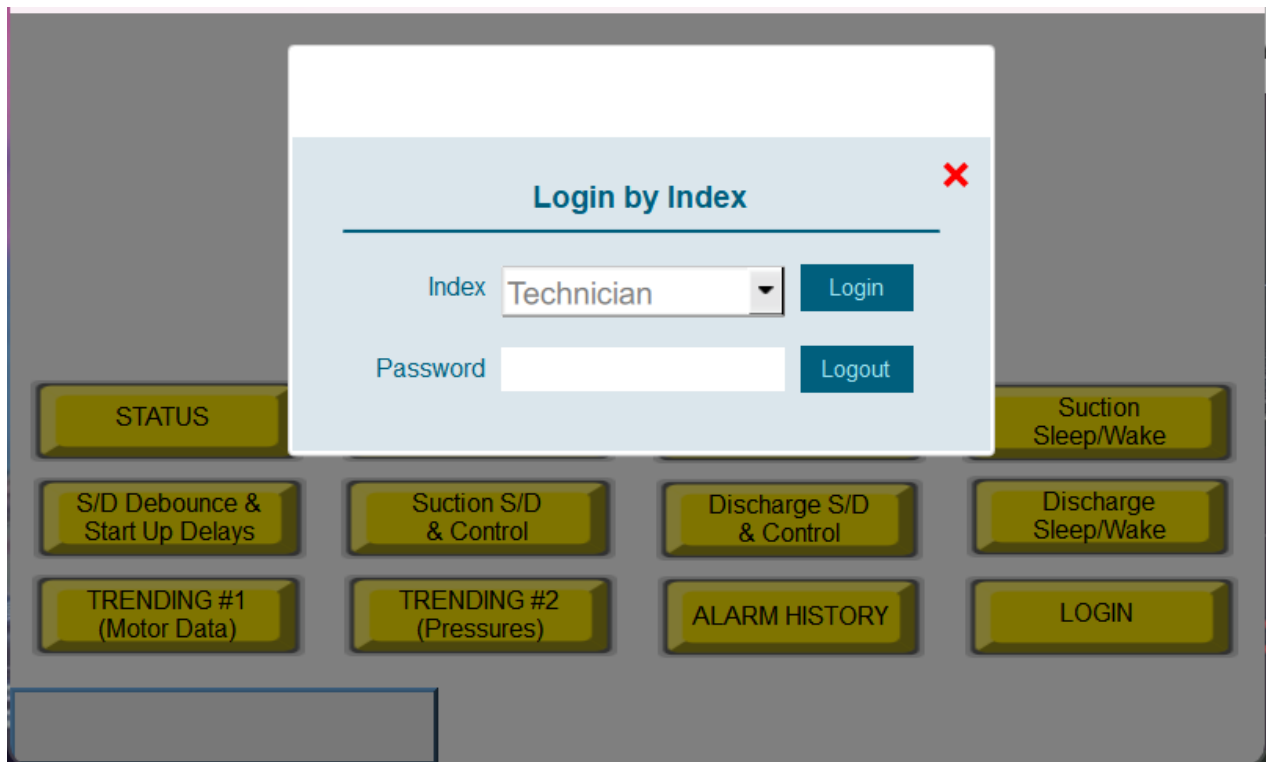


Figure 18 Alarm History page

- **Alarm History Display** – This display logs date and time of which the alarm occurred
- **RESET FAULT/ALARM** – Pressing this button will clear any active fault on the GCD.

Login Pop-up

Accessible from the main menu the login pop-up allows the technician to login in and enable the technician parameters. **The password for technician is 4444** the technician must enter the password in and press enter on the keypad after the password has been entered and the keypad is gone the login button must be pressed if it says succeeds under the password line the password was entered correctly and the technician mode is enabled. You may press the red X to close and then enter the parameters required.



The screenshot shows a 'Login by Index' pop-up window overlaid on a main menu. The pop-up has a title bar with a red 'X' close button. Inside, there is a section titled 'Login by Index' with a horizontal line below it. Below the line, there are two input fields: 'Index' with a dropdown menu showing 'Technician' and a 'Login' button, and 'Password' with a text input field and a 'Logout' button. The background menu consists of several yellow buttons with black text: 'STATUS', 'Suction Sleep/Wake', 'S/D Debounce & Start Up Delays', 'Suction S/D & Control', 'Discharge S/D & Control', 'Discharge Sleep/Wake', 'TRENDING #1 (Motor Data)', 'TRENDING #2 (Pressures)', 'ALARM HISTORY', and 'LOGIN'. There is also an empty rectangular box at the bottom left of the menu.

Figure 19 Login page

Troubleshooting

For common faults/alarms and their possible causes see table starting on page 13.

Faults/Alarms	Description	Possible Causes
F006 Imbalance or Input Phase Loss	Mains voltage imbalance too high or phase missing in the input power supply. Note: - If the motor is unloaded or operating with reduced load this fault may not occur. - Fault delay is set at parameter P0357. P0357 = 0 disables the fault. - If single-phase power supply is used, this fault must be disabled.	Phase missing at the inverter's input power supply. Input voltage imbalance >5 %. For the Frame Size E: Phase loss at L3/R or L3/S may cause F021 or F185. Phase loss at L3/T will cause F006. For frame sizes F and G: Pre-charge circuit fault.
F021 DC Link Undervoltage	DC Link undervoltage condition occurred	The input voltage is too low, and the DC bus voltage dropped below the minimum permitted value (monitor the value at parameter P0004): DC BUS Voltage < 446 V - For a 440-460 V input voltage (P0296 = 3). DC BUS Voltage < 487 V - For a 480 V input voltage (P0296 = 4). Phase loss in the input power supply. Pre-charge circuit failure. Parameter P0296 was set to a value above of the power supply rated voltage. When free spooling can increase P0169 and P0170 as needed depending on speed.

F022 DC Link Overvoltage	DC link overvoltage condition occurred.	The input voltage is too high, and the DC bus voltage surpassed the maximum permitted value: DC Bus Voltage > 400 V - For 220-230 V input models (P0296 = 0). DC Bus Voltage > 800 V - For 380-480 V input Inertia of the driven load is too high, or deceleration time is too short. Wrong settings for parameters P0151, or P0153, or P0185. Ensure P0185 and P0186 are set to 770 and 40 respectively. When free spooling can increase P0169 and P0170 as needed depending on speed.
F048 IGBT Overload Fault	An IGBTs overload fault occurred.	Inverter output current is too high.
F070 Overcurrent / Short-circuit	Overcurrent or short-circuit detected at the output, in the DC bus, or at the braking resistor.	Short-circuit between two motor phases. Short-circuit between the connection cables of the dynamic braking resistor. IGBT modules are shorted. Disconnect Motor try running VFD if issue persists replace inverter, if problem goes away check motor and cable for shorts perform a megger test.
F071 Output Overcurrent	The inverter output current was too high for too long.	Excessive load inertia or acceleration time too short. Settings of P0135 or P0169 and P0170 are too high
F072 Motor Overload	The motor overload protection operated. Note: It may be disabled by setting P0348 = 0 or 3	Settings of P0156, P0157 and P0158 are too low for the used motor. Motor shaft load is excessive.

F0150 Motor Overspeed	Overspeed fault. It is activated when the real speed exceeds the value of $P0134 \times (100 \% + P0132)$ for more than 20 ms.	Wrong settings of P0161 and/or P0162. Problem with the hoist-type load. Adjust P0132 higher or increase P01011.
A163/164 Break Detect AI1/AI2	It indicates that the AI current (4-20 mA or 20-4 mA) reference is out of the 4 to 20 mA range.	Broken AI cable. Bad contact at the signal connection to the terminal strip. Blown control fuse on terminal strip for AI circuit. If there is no AI going to that input set the corresponding input P0233-AI1 or P0238-AI2 to 0.
A181 Invalid Clock Value	Invalid clock value alarm	It is necessary to set date and time at parameters P0194 to P0199. (Try this first if not then check battery) Keypad battery is discharged, defective, or not installed.
F185 Pre-charge Contactor Fault	It indicates fault at the pre-charge contactor	Pre-charge contactor defect. Open command fuse. Phase loss in the input L1/R or L2/S. P0355 = 1 (Incorrect setting for model's frame E supplied by the DC link. For those models, the setting should be P0355 = 0.

For CFW11 Alarm & Fault List refer to page 6-2 of the CFW-11 User Manual Provided

Faults/Alarms

F750 – Panel ESD Shutdown	F751 – High Discharge Pressure
Shutdown – Digital Input	
F752 – High Discharge Temp Shutdown	F753 – Scrubber High Level #1
Shutdown	
F754 – High Suction Pressure Shutdown	F755 – Scrubber High Level #2
Shutdown	
F757 – Low Suction Pressure Shutdown – Digital Input	F758 – Low Oil Pressure
Shutdown	
F759 – Shutdown #1	F760 – Shutdown #2
F761 – Low Pressure Differential	F762 – Low Suction Pressure
Shutdown – Transmitter (AI2)	
F763 – High Discharge Pressure Shutdown – Transmitter (AI1)	A764 – Both PID's Enabled
A765 – Suct PID Sleep	A766 – Dis PID Sleep

NOTE:

- Red Button on Keypad is utilized to stop VFD in local or reset alarms/faults
- Green Button on Keypad is utilized to start VFD in local

Digital Inputs

DI1 – Run/Stop
DI2 – Manual/Auto
DI3 – Local/Remote
DI4 – Shutdown#1
DI5 – Shutdown #2
DI6 – Low Pressure Differential
DI9 – Low Suction Pressure
DI10 – Low Oil Pressure
DI11 – High Discharge Pressure
DI12 – High Discharge Temp
DI13 – Scrubber High Level#1
DI14 – High Suction Pressure
DI15 – Scrubber High Level #2
DI16 – Panel ESD

Analog Input / Output Configuration (Dipswitch's Located at top of control board)

S1.1 (Analog Output 1): Dip Switch Off - 4-20mA, Dip Switch On: 0-10vdc

S1.2 (Analog Output 2): Dip Switch Off - 4-20mA, Dip Switch On: 0-10vdc

S1.3 (Analog Input 2): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

S1.4 (Analog Input 1): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

Technical Support

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